

Proving triangles similar



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To prove: $\triangle ABC \sim \triangle XYZ$

Statements	Reasons
1. $\frac{AC}{XZ} = \frac{BC}{YZ}$	Given
2. $\angle ACB \cong \angle XZY$	Given
3. $\triangle ABC \sim \triangle XYZ$	Side-angle-side similarity (SAS $\sim$)

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To prove: The two triangles are similar.

Statements	Reasons
1. $\frac{AC}{YZ} = \frac{AB}{XZ} = \frac{CB}{XY}$	Given
2. $\triangle ABC \sim \triangle ZYX$	Side-side-side similarity (SSS $\sim$)

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To prove: The two triangles are similar.

Statements	Reasons
1. $\angle IHJ \cong \angle LKJ$	Given
2. $\angle IJH \cong \angle LJK$	Vertical angles congruence theorem
3. $\triangle HIJ \sim \triangle K LJ$	Angle-angle similarity (AA $\sim$)

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To prove: The two triangles are similar.

Statements	Reasons
1. $\frac{AC}{BC} = \frac{EC}{DC}$	Given
2. $\angle ACE \cong \angle BCD$	Reflexive property of congruence
3. $\triangle ACE \sim \triangle BCD$	Side-angle-side similarity (SAS $\sim$)

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To prove: The two triangles are similar.

Statements	Reasons
1. $\overline{AB} \parallel \overline{DE}$	Given
2. $\angle BAC \cong \angle EDC$	Alternate interior angles theorem
3. $\angle DCE \cong \angle ACB$	Vertical angles congruence theorem
4. $\triangle ABC \sim \triangle DEC$	Angle-angle similarity (AA $\sim$)

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To prove: $\triangle ACE \sim \triangle BCD$

Statements	Reasons
1. $\angle CBD \cong \angle CAE$	Given
2. $\angle BCD \cong \angle ACE$	Reflexive Property of Equality
3. $\triangle ACE \sim \triangle BCD$	Angle-angle similarity (AA $\sim$)

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To prove: $\triangle ABC \sim \triangle DEC$

Statements	Reasons
1. $\frac{BC}{CE} = \frac{AC}{CD}$	Given
2. $\angle BCA \cong \angle ECD$	Vertical angles congruence theorem
3. $\triangle ABC \sim \triangle DEC$	Side-angle-side similarity (SAS $\sim$)

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To prove: $\triangle ABC \sim \triangle DEC$

Statements	Reasons
1. $\angle BAC \cong \angle EDC$	Given
2. $\angle BCA \cong \angle ECD$	Vertical angles congruence theorem
3. $\triangle ABC \sim \triangle DEC$	Angle-angle similarity (AA $\sim$)



Additional questions

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- a True
- b False
- c False
- d False
- e True

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- a $\triangle AYP$ is not similar to $\triangle BOX$ because $\angle A \not\cong \angle B$.
- b $\triangle JKL \sim \triangle JMN$ by the SAS similarity theorem.
- c $\triangle UVW \sim \triangle FGH$ by the SAS similarity theorem.
- d $\triangle ZYG \sim \triangle BRP$ by the SSS similarity theorem.

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- a Since $\angle A \cong \angle F$ and $\angle B \cong \angle C$, $\triangle ABC \sim \triangle FCE$ by the AA similarity theorem.
- b Since $\angle MVL \cong \angle UVT$ and $\frac{MV}{VT} = \frac{LV}{VU}$, $\triangle MVL \sim \triangle TVU$ by the SAS similarity theorem.
- c Since $\angle U \cong \angle F$ and $\frac{UW}{HF} = \frac{UV}{FG}$, $\triangle UVW \sim \triangle FGH$ by the SAS similarity theorem.
- d Since $\frac{FR}{FE} = \frac{FS}{FD} = \frac{SR}{DE}$, $\triangle FSR \sim \triangle FDE$ by the SSS similarity theorem.

12 $x = 11$